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AI Governance Portfolio — Artifact #1

**NIST AI RMF Risk Assessment: FraudShield AI**

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## EXECUTIVE SUMMARY

I applied the NIST AI Risk Management Framework (AI RMF 1.0) to a fintech fraud detection system with LLM-generated fraud explanations. The assessment identified **15 AI risks** across all four RMF functions — Govern, Map, Measure, Manage — and produced a decision-ready recommendation to pause deployment pending remediation.

## KEY FINDINGS

- **9 of 15 risks (60.0%)** classified as Critical
- **Highest single exposure:** System unavailability during peak hours (score 25 of 25)
- **No Medium-tier risks** — distribution is inverted toward Critical, indicating systemic rather than isolated governance gaps
- **Category concentration:** Technical risks dominate count (7); Operational and Ethical risks carry highest individual severity

## ARTIFACTS PRODUCED

- Risk register (15 scored risks mapped to NIST functions)
- Heat map with quadrant analysis
- Category bar chart (min/average/max by risk type)
- Risk-level pie chart (colorblind-safe Okabe-Ito palette)
- Ranked priority list

## SKILLS DEMONSTRATED

- NIST AI RMF 1.0 implementation from framework to decision
- Quantitative risk scoring and tiering methodology
- GenAI-specific risk identification (prompt injection, model memorization, vendor deprecation — identified as extension candidates)
- Data analysis and visualization (pandas, matplotlib)
- Executive communication of technical risk

## EXTENSION CANDIDATES

Beyond the required assessment, I identified three GenAI-specific failure modes not covered by the base exercise:

- **Prompt injection** via LLM-generated fraud explanations
- **Training data memorization** leaking PII in generated text
- **Vendor model deprecation** forcing unplanned migration

These demonstrate where the NIST AI RMF requires extension for LLM-based system